

BIRLA INSTITUTE OF TECHNOLOGY MESRA, RANCHI
DEPARTMENT OF MATHEMATICS

MA205 Discrete Mathematics: (MO-2022)

Assignment Questions (Instructor: Dr. Prabal Datta)

1. Find the values of r_1, r_2 and r_3 in the recurrence relation

$$a_n = r_1 a_{n-1} + r_2 a_{n-2} + r_3 a_{n-3}, n \geq 3$$

if the general solution is $a_n = A(-1)^n + Bn(-1)^n + C3^n$.

2. Find r_1, r_2 and r_3 if the recurrence relation

$$a_n + r_1 a_{n-1} + r_2 a_{n-2} + r_3 a_{n-3} = 0, n \geq 3$$

has a general solution of the form $a_n = u2^n + vn2^n + w3^n$.

3. Consider the $(3, 5)$ group encoding function $e : B^3 \rightarrow B^5$ defined by

$e(000) = 00000, e(001) = 00110, e(010) = 01001, e(011) = 01111, e(100) = 10011, e(101) = 10101, e(110) = 11010, e(111) = 11100$.

Decode the following words relative to a maximum likelihood decoding function:

(1) 11001

(2) 01010

(3) 00111

4. Give an example of a regular, connected graph on six vertices that is not complete?
5. Give an example of a graph that is regular, but not complete, with each vertex having degree three.